

Ali Aghababaei

AI Engineer | LLM & Agentic Systems

Orientation Year Visa (zoekjaar) | authorized to work in the Netherlands without sponsorship

Email: ali.aghababaei96@gmail.com

Phone: +31 6 83 35 97 62

Location: Rotterdam, Netherlands

LinkedIn: linkedin.com/in/aliaghababaei

Portfolio: aghababaeiali.github.io

GitHub: github.com/aghababaeiali

Professional Summary

Applied AI Engineer building production LLM systems, from orchestrated LangGraph pipelines to genuinely agentic architectures with permission-scoped tool access. Ships end to end on AWS and Azure with Terraform and CI/CD, and reports honest metrics. First co-author on a 2025 arXiv paper on NLP explainability. MSc ICT, University of Padova, 110/110.

Experience

Independent AI Engineering, Rotterdam, Netherlands

Agentic Systems & LLM Infrastructure · Aug 2025 – Present

- Building and deploying production-shaped LLM systems end to end: agent architecture, evaluation, cloud infrastructure, and CI/CD. Selected work below.

Research Assistant, University of Padova, Italy

Explainable AI & Transformer Evaluation · Jun 2024 – Dec 2024 · Part-time

Recovering which words drove a transformer classifier's decision, and establishing which attribution method could be trusted to do it.

- Benchmarked 4 attribution methods (Integrated Gradients, Sequential IG, DeepLIFT, GradientSHAP) across 4 datasets; selected IG on a performance-versus-cost basis
- Optimized IG hyperparameters against defined quality metrics: integration steps and baseline embedding choice
- Structured human evaluation with 3 expert psychologists surfaced a **20% false-attribution rate**; reported the limitation rather than the headline number
- Fine-tuned BERT and RoBERTa with controlled overfitting for class-distinctive token attribution; built a SpaCy pipeline resolving subword tokenization and negation alignment for readable word-level output. First co-author, arXiv 2025 ([code](#))

Test Engineer, Mazand Phase Industrial Power, Iran

Test Engineering & Data Analysis · Nov 2019 – Mar 2022 · Full-time

- Replaced manual Excel reporting with a Python/SQLite pipeline and automated statistical process control charts that caught supplier quality drift before specification failures
- Built unsupervised anomaly detection (Isolation Forest) over engineered features, flagging units that deviated from the historical passing population despite individually in-spec readings

Projects

Dealership Agent (LangGraph, MCP, AWS, PostgreSQL)

[GitHub](#)

Aug 2026

A customer service agent for a used-car dealership. The design question was not whether it could find a car, but whether the LLM could be tricked into reading another customer's data.

- Designed a multi-agent system where authorization is structural, not prompted: sub-agents split by tool permission scope, identity never enters tool schemas, and PostgreSQL row-level security enforces the boundary at the data layer
- Built a 71-case labelled eval set for hallucinated action claims; a two-stage verifier raised accuracy **0.690 to 0.901**
- Deployed to ECS Fargate, RDS/pgvector and ALB via Terraform and OIDC-secured CI; live verification surfaced two production-only bugs invisible to local testing

Property Insights Assistant (LangGraph, PostgreSQL, Azure)

[GitHub](#)

Jul 2026

Natural-language questions over property data, with a model surfacing bookings likely to cancel.

- Built an 8-node LangGraph workflow with SQL, RAG, and risk routing; added deterministic answer grounding, sampled LLM-as-judge, and a human-review queue feeding a regression suite
- Removed target leakage from the cancellation-risk model, compared 4 model families via cross-validation, reported an honest **0.706 ROC-AUC**
- Migrated to Azure end-to-end with Terraform and OIDC-secured CI/CD; Application Insights tracing shows **p50 29ms** on the risk endpoint

arXiv Search Assistant (LangGraph, ChromaDB, RAGAS)

[Live](#) | [GitHub](#)

Apr 2026

A pipeline that searches arXiv and synthesises cited answers, flagging where sources contradict.

- Built an 8-node LangGraph system: query reformulation loop, session-isolated retrieval with per-paper re-ranking, contradiction detection, grounded synthesis with citations
- Evaluated with RAGAS across 20 queries: **Faithfulness 0.921**; deployed live on Hugging Face Spaces

Education

M.Sc. in ICT for Internet and Multimedia, University of Padova, Italy

Sep 2022 – Jul 2025 | Grade: 110/110

Erasmus Exchange, Warsaw University of Technology, Poland

Oct 2023 – Feb 2024 | Grade: 4.75/5.0

B.Sc. in Electrical Engineering, Ferdowsi University of Mashhad, Iran

Sep 2014 – Sep 2019

Technical Skills

Languages: Python, SQL

ML & AI: PyTorch, Hugging Face Transformers, Scikit-learn, NumPy, Pandas, Anomaly Detection, Feature Engineering

NLP & GenAI: LangGraph, Agentic AI, RAG, LLMs, Model Context Protocol (MCP), Prompt Engineering, Information Retrieval, BERT, RoBERTa, LangChain, Vector Search, pgvector, ChromaDB, sentence-transformers, LoRA/PEFT, SpaCy, Groq, AWS Bedrock

Evaluation: RAGAS, LLM-as-judge, Explainable AI (XAI), Integrated Gradients, Captum, SHAP, Data Curation

Cloud & Infrastructure: AWS (ECS Fargate, RDS, Bedrock, ALB, ECR, IAM, CloudWatch), Azure (Container Apps, OpenAI, AI Search, ML), Terraform, CI/CD (GitHub Actions, OIDC), Docker, FastAPI, REST APIs, PostgreSQL, pytest, MLflow, Application Insights

Publications

Application of Integrated Gradients Explainability to Sociopsychological Semantic Markers

arXiv preprint (2025) · arxiv.org/abs/2503.04989

Artificial Intelligence in Agro-Food Systems: From Farm to Fork

Foods Journal, MDPI (2025) · doi.org/10.3390/foods14030411

Languages

English (C2, IELTS 8.5) Persian (Native) Italian (A2) Dutch (beginner, learning)